

Carson Godwin

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PROFESSIONAL EXPERIENCE

Software Engineer | Kimley-Horn

March 2025 - Present

- Resolved a production outage on the core analytics engine via emergency patch, converting O(n) lookups to O(1) indexing and parallelizing async REST API calls, cutting load times by 50%+ and backend latency by 20%.
- Unblocked the processing of annual renewals by engineering a fault-tolerant Python batch pipeline across Azure Cosmos DB, updating 30,000+ documents to integrate 1,500 active subscriptions into the automated billing cycle.
- Built a self-service document management portal on Azure Blob Storage, eliminating developer dependency for stakeholder-managed content and reducing update time from 15 minutes to under 10 seconds.
- Refactored the Project Notes module full-stack, replacing collection-scan queries with Cosmos DB point reads by document ID, eliminating unnecessary RU consumption and reducing query overhead across client workloads.

Backend Engineer & Co-Founder | Impact Point

August 2025 - Present

Joined as technical co-founder to scale live multiplayer game with 280,000+ players on Steam

- Enhanced C# bot AI decision loops using static registries and zero-allocation targeting, eliminating over 400 frame-by-frame memory allocations per second to resolve severe 200ms+ GC-induced network latency spikes.
- Orchestrated a zero-downtime platform migration from Steamworks to PlayFab, preserving live player economy data and progression while enabling cross-platform mobile expansion.
- Optimized map loading pipeline using asynchronous asset streaming (lazy loading), reducing load times by 95% and eliminating main-thread blocking.
- Architected event-driven Azure Cloud Functions to validate and persist end-of-match player stats to PlayFab, moving win streak logic server-side to prevent client manipulation of progression data.

Software Engineer Intern | Infosys

May - July 2024

- Built a full-stack AI documentation assistant (Flask, Azure OpenAI) that let non-technical stakeholders generate structured client deliverables without developer involvement, reducing creation time from 3-4 hours to under 5 minutes.
- Engineered the backend logic using LangChain, Pandas, and Azure OpenAI to parse Excel requirements and generate reliable, strictly formatted outputs via few-shot prompting.

Machine Learning Research Assistant | College of Science and Engineering

September 2022 - January 2024

- First author on a published IEEE paper regarding search personalization; engineered a custom Python automation suite (Selenium, BeautifulSoup) to harvest 10,000+ search results for analysis.
- Built a "virtual personality" simulation tool to mimic organic user behavior, generating a dataset used to train machine learning models on search engine privacy algorithms.

PROJECTS

Karto | iOS Social Network | Lead Developer

March 2023 - February 2025

2nd Place, EIBF Entrepreneurship Competition (Edinburgh, Scotland) Lightbulb Prize & Microfinance Award recipient

- Architected a location-based social platform from concept to App Store, implementing radius-based geospatial queries with Google Maps API to render real-time user feeds directly on an interactive map.
- Reduced Firebase costs by 30% by implementing a local device caching layer that intercepted redundant database reads, serving repeated queries from cache while maintaining real-time sync integrity.
- Built an event-driven push notification system via Google Cloud Functions, triggering real-time alerts on social interactions (likes, friend requests) and driving a 40% increase in same-day app opens following a notification.

TECHNICAL SKILLS & TOOLS

Languages: Python, C#, TypeScript, JavaScript, Dart, HTML, CSS

Frameworks & Tools: .NET, Angular, Unity, Flutter, Git, CI/CD (Azure DevOps)

Databases & Services: Cosmos DB, Azure, PlayFab, Firebase, Google Cloud

EDUCATION

University of North Carolina Wilmington (UNCW)

December 2024

B.S., Computer Science, Business Concentration, Dean's List

Association for Computing Machinery (ACM) - Chapter Founder and President